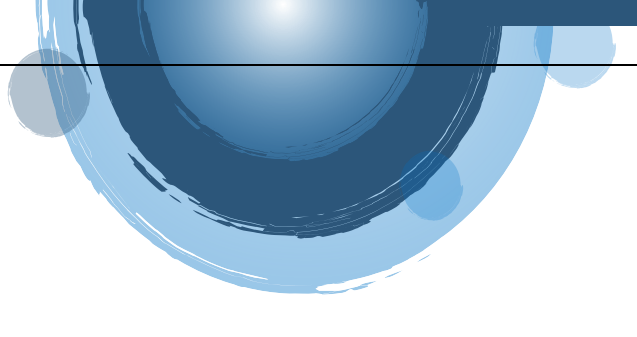




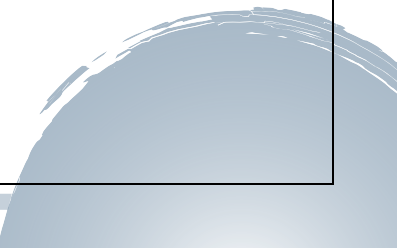
Cyber Security Internship Report

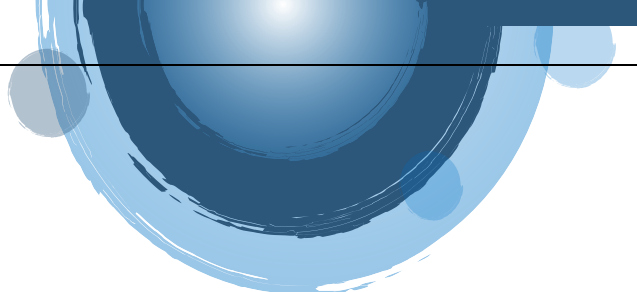
Cyber Security Internship – Task 11

Phishing Attack Simulation & Detection Report

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Date : 02 / 02 / 2026



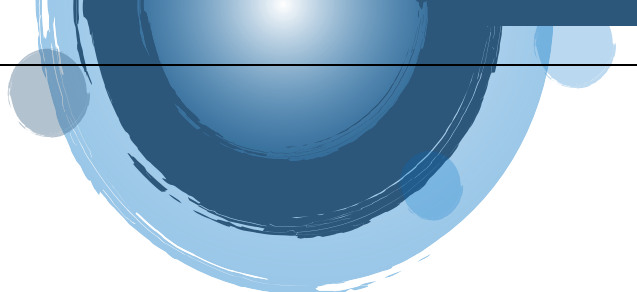


1. Introduction

Phishing is a type of social engineering attack in which attackers attempt to trick users into revealing sensitive information such as usernames, passwords, credit card details, or personal data by pretending to be a trusted entity. These attacks are commonly carried out through fraudulent emails, websites, or messages that appear legitimate.

Phishing is one of the most common causes of data breaches and financial loss worldwide. Organizations conduct phishing simulations to educate users, test awareness, and improve cybersecurity defenses.

In this task, a phishing attack simulation was performed using GoPhish to understand how attackers craft phishing emails and how such attacks can be detected and prevented.



2. Objectives

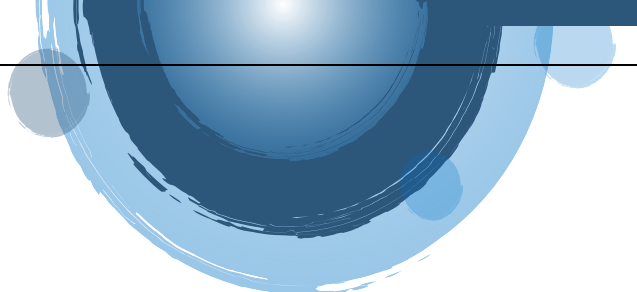
The main objectives of this task were:

- Understand the concept of phishing attacks
- Learn how phishing emails are created
- Simulate a phishing campaign using GoPhish
- Create a fake landing/login page
- Send test phishing emails
- Track user interactions
- Identify phishing indicators
- Learn detection and prevention methods

3. Tools Used

The following tools were used during the simulation:

- GoPhish – Open-source phishing simulation framework
 - Kali Linux – Testing environment
 - Firefox Browser – Accessing admin dashboard
 - Localhost server – Campaign testing
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4. Setup Process (Installation Steps)

Step 1: Downloaded GoPhish from the official website

Step 2: Extracted the zip file

Step 3: Gave execution permission

```
chmod +x gophish
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Step 4: Started the server

```
./gophish
```

Step 5: Accessed admin panel at

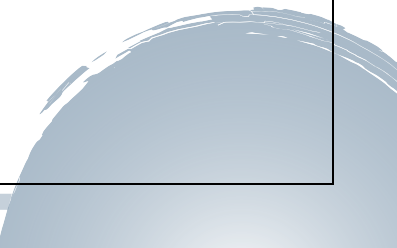
<https://127.0.0.1:3333>

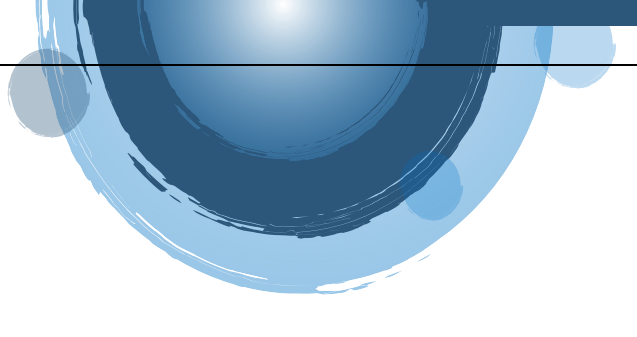
Step 6: Logged in with default credentials

5. Email Template Creation

An email template was created to simulate a real phishing message.

The email contained:

- Urgent language
 - Fake account warning
 - Clickable link
 - IT support impersonation
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Sample template:

Subject:

Important: Account Verification Required

Body:

Dear User,

Your account will be suspended.

Please verify immediately using the link below.

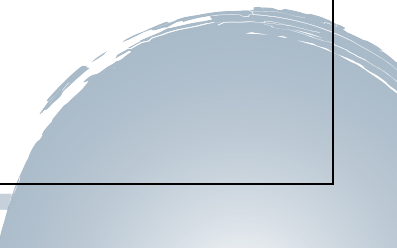
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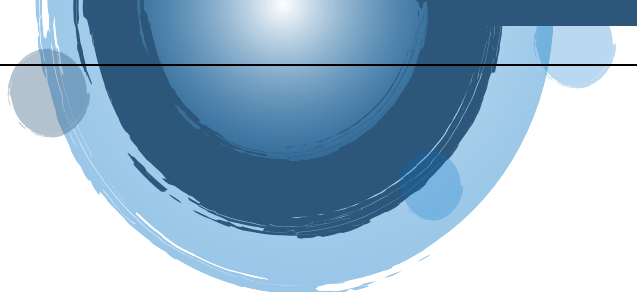
This template tricks users into clicking the malicious link.

6. Landing Page Creation

A fake login page was created to capture user credentials.

The landing page contained:

- Username field
 - Password field
 - Submit button
- 



Credential capture was enabled to record entered information.

Purpose:

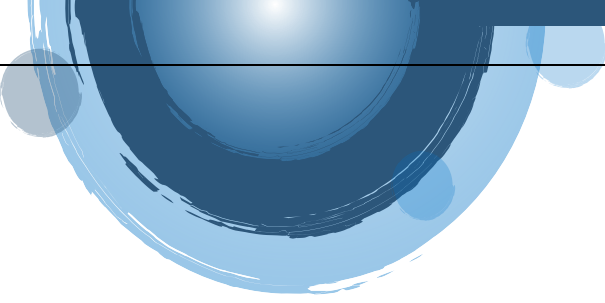
To simulate how attackers steal login credentials.

7. Campaign Execution

The phishing campaign was launched with the following steps:

- Created user group with test email addresses
- Configured sending profile
- Selected email template
- Selected landing page
- Started campaign

After launching, emails were sent to the target users and their activities were tracked.



8. Results (Screenshots)

The campaign dashboard showed:

- Number of emails sent
- Email opens
- Link clicks
- Credentials captured

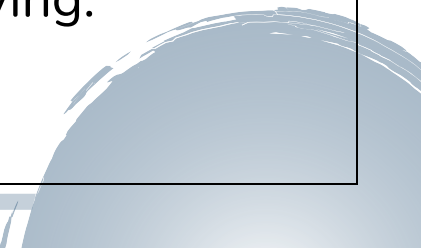
Results helped analyze user behavior and vulnerability.

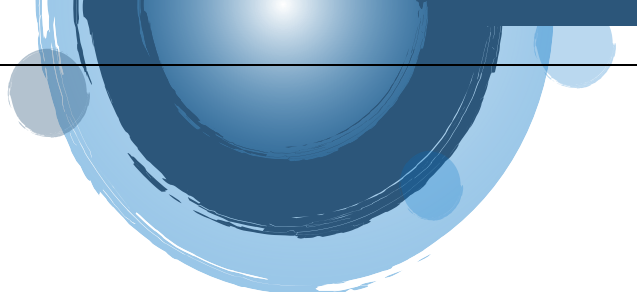
Example observations:

- Users clicked suspicious link
 - Some users entered credentials
 - Demonstrates real-world risk
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9. Detection Techniques

Phishing attacks can be detected by observing:

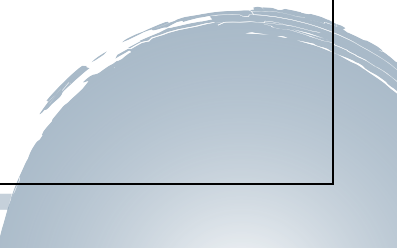


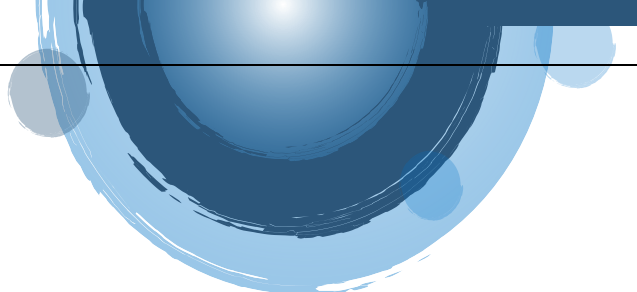
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- Suspicious sender email
 - Spelling/grammar mistakes
 - Urgent or threatening language
 - Unknown links
 - Requests for passwords
 - Fake domain names
 - HTTPS mismatch

Users should verify emails before clicking links.

10. Prevention Methods

The following measures help prevent phishing attacks:

- User awareness training
 - Email spam filters
 - Two-factor authentication (2FA)
 - Avoid clicking unknown links
 - Verify website URLs
 - Anti-phishing tools
 - Strong passwords
 - Regular security audits
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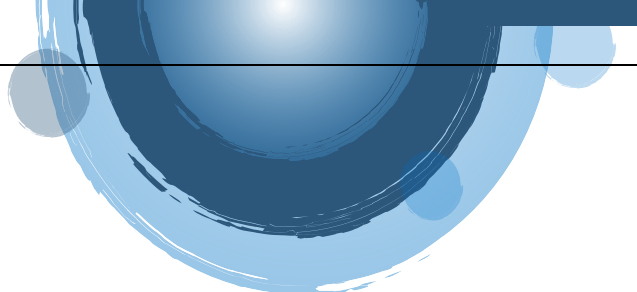


Prevention reduces the risk of credential theft and data breaches.

11. Learning Outcomes

After completing this task, the following skills were gained:

- Practical knowledge of phishing attacks
 - Experience with GoPhish tool
 - Understanding of social engineering tactics
 - Ability to simulate phishing campaigns
 - Knowledge of detecting phishing emails
 - Awareness of cybersecurity best practices
 - Hands-on experience with attack simulation and defense
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12. Conclusion

This phishing attack simulation provided a practical understanding of how attackers design fraudulent emails and steal sensitive information. The exercise demonstrated that users can easily fall victim to social engineering attacks if proper awareness is not maintained.

Using GoPhish helped in understanding both the attacker's perspective and defensive strategies. Implementing proper detection techniques and preventive measures is essential to protect organizations from phishing threats.

Overall, this task improved my knowledge of cybersecurity awareness, attack simulation, and phishing detection.
